Nathan Espejo

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Education

Western University — London, ON
Bachelor of Engineering Science in Software Engineering, Dean's Honour List (2024, 2025) — *Aug 2024 - May 2028*

Experience

VP of Data/ML — May 2026 - Aug 2026
3D Western — *London, ON*

- Co-led two data developers on the print-safety dataset and benchmark, tracking work on a GitHub project board.
- Built a labelled dataset from 4,620 scraped 3D models, rendered in Blender with labels checked by a VLM.
- Benchmarked small vision models as a binary safe-or-flag gate: 84.2% best agreement with VLM-checked labels.
- On the data team since Sep 2025, first working on the 3D model scraping and the STL rendering pipeline.

Software Engineering Intern — May 2026 - Aug 2026
TeraGo — *Markham, ON*

- Built a prototype pipeline that uses a locally hosted LLM (Ollama) to turn unstructured emails into structured tickets, working end to end.
- Built a Python RF channel-planning tool with a command-line and web interface that shortlists clear frequencies for fixed-wireless links, with an automated validation suite.
- Deployed an internal RAG assistant that answers engineers' technical questions from company documentation.
- Researched and proposed a self-hosted LLM platform integrated with internal tools.

Projects

Campus | *Python, FastAPI, SQLite, React, TypeScript, Docker* — Aug 2026 - Present

- Built an AI agent that reads all six of my courses, syncing them from the LMS into SQLite with Playwright and running on any OpenAI-compatible endpoint, local or hosted.
- Designed search over course files that is lexical by default with optional embeddings and reranking, and added a lexical SQL prefilter for exact phrases that scored below the semantic cutoff.
- Wrote a 20-tool agent harness with an audit log for every AI write and a bounded tool loop, with 403 backend tests in CI.
- Use it across six courses for deadline lookups, studying from course slides, and reviewing drafts against briefs.

Homelab & NixOS | *NixOS, Hermes Agent, Docker, Tailscale* — May 2026 - Present

- Defined a laptop and home server in one git-tracked NixOS config, with 35 containers behind Tailscale.
- Run one Hermes Agent profile per host that makes changes as config edits, across 411 sessions since June 2026.
- Gave each agent its own level of autonomy: the server profile waits for approval on a full diff, the laptop profile commits and asks me to test, and both rebuild through a host-checked guard.
- Grew a git-versioned library of 212 agent skills run on low-cost models picked per task, pruned weekly by a curator that keeps backups of removed skills.

AWARE (Hack the 6ix, solo) | *Python, YOLOv8, YAMNet, ONNX, Qualcomm QNN* — Jul 2026

- Built a home-awareness device running YOLOv8 and YAMNet on an $80 Arduino UNO Q, no cloud inference.
- Used an on-device LLM to compile commands like "when glass breaks after 10pm alert me" into deterministic rules, checked every 500 ms with no LLM in the loop.

Hackathon Awards

8 awards in 22 hackathons: NSBEHacks 3rd place, 1st for Cohere (Careerly) | Hack Western Best AI Application with Cloudflare (Auralis) | TerraHacks Best Use of MongoDB (Lumen) | Spark 2026 3rd (CareBridge, contributor)

Technical Skills

AI: LLM agents, tool calling, RAG, embeddings and reranking, MCP, VLMs, model evaluation, llama.cpp, Ollama, PyTorch, YOLOv8, ONNX
Languages: Python, TypeScript, C#, C++, SQL, Bash
Web & Infra: React, FastAPI, Next.js, NixOS, Linux, Docker, Tailscale, Caddy, Git, GitHub Actions
Tools: Hermes Agent, Blender